Ousama Alabdullah

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EDUCATION

Toronto Metropolitan University **Toronto, Ontario**
B.Eng. in Software Engineering *May 2026*
Relevant Coursework: Distributed Cloud Computing, Algorithms & Data Structures, Database Systems (SQL, transactions, JDBC), Operating Systems (processes, threads, IPC), Computer Networks, Network Security, Object-Oriented Engineering Analysis & Design, Software Project Management, Machine Learning, Capstone Design Project
Certifications: AWS Certified Cloud Practitioner, AWS Certified AI Practitioner

EXPERIENCE

Ministry of Transportation Ontario **Toronto, ON**
Engineering Intern (Software / Data) *May 2024 – Aug 2025*

- Built automated **ETL pipelines in Python** that pulled traffic data from HERE, TomTom, and Waze across 13 routes, **cutting manual processing time by 45%**.
- Wrote a **Python backend service** that read XML data from the HERE API and used Shapely and GeoPandas to clean and validate map data before saving it.
- Managed **Oracle / SQL** databases in production, keeping data clean, queries fast, and access controlled through code reviews.
- Shipped an internal **web app** with an interactive map for filtering systems by area or highway, **automating 50% of routine tasks** and cutting lookup time by 30%.
- Set up **CI/CD pipelines** for version control, testing, and deployment, making releases 30% faster.
- Used **AI tools (Claude, Copilot)** to speed up coding, write docs, and run QA checks, with peer review and tests on every AI-assisted change before merging.

PROJECTS

Break Even, Full-Stack Finance Application [Link] *Ruby on Rails, PostgreSQL, React, JWT, REST API*

- Built a **REST API in Ruby on Rails** with **PostgreSQL** to handle users, budgets, expenses, and daily ledger entries, with logic for daily budgeting, automatic rollover of unspent money, and full history tracking.
- Designed a **daily ledger system** that pre-computes balances for **fast reads** and recalculates them safely whenever a past expense or budget is edited.
- Built a **React frontend** that talks to the API, with instant UI updates and **form validation** on both the client and server for a smooth experience.
- Added **JWT login** with **Supabase and Google OAuth**, and used solid backend practices like storing money as integers, handling time zones, indexing key queries, and writing **automated tests**.

NAO Interactive Robot System [Link] *Python, Flask, NAOqi, OpenCV, OpenAI API, WebSockets*

- Built a real-time **voice chat pipeline** where the NAO robot sends audio over a **WebSocket** to a **Flask server**, which sends it to the **OpenAI API** and streams the spoken reply back to the robot.
- Designed the **Flask service** to be stateless, with **retries on failed API calls** and a fallback when the OpenAI API is slow or down.
- Added **face and object recognition** using a **CNN model** so the robot can recognize people and respond based on what it sees.
- Coordinated audio, vision, and motor control on **separate threads** through the **NAOqi API** so nothing blocks the main loop.

Document Scanner & OCR System [Link] *Python, OpenCV, Tesseract, NumPy*

- Built a **document scanning pipeline** that finds the edges of a document in a photo, straightens it into a flat top-down view, and pulls out the text using **Tesseract OCR**.
- Improved text accuracy in poor lighting and bad angles using **denoising, CLAHE contrast, and adaptive thresholding** to clean up the image before reading it.

Machine Learning Algorithms from Scratch (ELE888 Coursework) [Link] *Python, NumPy, Matplotlib*

- Built a **Bayesian classifier** on the IRIS dataset using posterior probabilities, with optimal decision boundaries for both equal and weighted misclassification costs.
- Implemented a **perceptron with gradient descent** to classify IRIS data, comparing accuracy across training splits and learning rates.
- Built a **multi-layer neural network with backpropagation** from scratch to solve the XOR problem, plotting learning curves and decision surfaces in input and hidden-layer space.
- Implemented **K-means clustering** to find dominant colors in an image, using the **Xie-Beni index** to compare cluster quality across initial states.

TECHNICAL SKILLS

Languages & Backend: Python, Java, C, JavaScript, TypeScript, Ruby, SQL, Spring Boot, Ruby on Rails, Flask, Node.js, RESTful APIs, JWT, OAuth, microservices, caching, ETL pipelines
Data & Cloud: PostgreSQL, Oracle, MongoDB, MySQL, Redis, AWS, GCP, Docker, Kubernetes, Git, CI/CD, GitHub Actions,
AI Tools: Claude Code, Cursor, GitHub Copilot, OpenAI API, Anthropic API, Loveable